

[Aim: 100/100 in Maths]

आरंभ

(A.P.)

ARITHMETIC PROGRESSION

Lecture #1

What is an A.P.?

↳ A sequence in which each term is obtained by adding a fixed number to the previous term.

common
difference
(d)

AP: $a_1, a_2, a_3, a_4, a_5, a_6, a_7, \dots, a_n$
 $3, 5, 7, 9, 11, 13, 15, \dots$
Arrows show the progression with labels: a , $+2$, $+2(d)$, $+2(d)$, $+2$.

fix no.
(d)
2

common diff (d) = $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = a_5 - a_4 = \dots = 2$
[अगला - पिछला] ← AP की पहचान $\Rightarrow$ all c.d. must be same

General form of AP: $a, a+d, a+2d, a+3d, a+4d, a+5d, \dots$

#K3B

$$a_1 = a$$

$$a_2 = a + 1d = a + (2-1)d$$

$$a_3 = a + 2d = a + (3-1)d$$

$$a_4 = a + 3d = a + (4-1)d$$

$$\vdots$$
$$a_{10} = a + (10-1)d$$

$$\vdots$$
$$a_{100} = a + (100-1)d$$

$$\vdots$$
$$a_{1000} = a + (1000-1)d$$

$$a_n = a + (n-1)d$$

← n^{th} term of AP

n^{th} term of AP $\Rightarrow a + (n-1)d$

LP: for an AP: $\textcircled{2}, 4, 6, 8, \dots$, find $\textcircled{101^{\text{th}}}$ term?

~~$a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}$
 $2, 4, 6, 8, 10, 12, 14, 16, 18, 20,$
 $+2, +2, +2$~~

which term

$n=101$

$$a_n = a + (n-1)d$$

$$a_{\underline{101}} = a + (\underline{101}-1)d$$

$$a_{101} = 2 + (100)(2)$$

$$= 2 + 200$$

$$\boxed{a_{101} = 202} \quad \checkmark$$

$$d = 4 - 2 \rightarrow \textcircled{2}$$

#LP: Find out which of the following sequence are arithmetic progressions. For those which are arithmetic progressions, find out the common difference.

i. $\overset{a_1}{3}, \overset{a_2}{6}, \overset{a_3}{12}, \overset{a_4}{24} \rightarrow \text{A.P.??}$

$$d_1 = a_2 - a_1 = 6 - 3 \Rightarrow \textcircled{3}$$

$$d_2 = a_3 - a_2 = 12 - 6 \Rightarrow \textcircled{6}$$

$$d_3 = a_4 - a_3 =$$

Since

$$d_1 \neq d_2$$

Not an AP!!

ii. $\overset{a_1}{0}, \overset{a_2}{-4}, \overset{a_3}{-8}, \overset{a_4}{-12}, \dots\dots\dots$

$$d_1 = a_2 - a_1 = -4 - 0 \Rightarrow \textcircled{-4}$$

$$d_2 = a_3 - a_2 = (-8) - (-4) \Rightarrow -8 + 4 \Rightarrow \textcircled{-4}$$

$$d_3 = a_4 - a_3 = (-12) - (-8) \Rightarrow -12 + 8 \Rightarrow \textcircled{-4}$$

Since $d_1 = d_2 = d_3$ Yes it's AP!!

$$\therefore \boxed{c.d(d) = -4} \text{ Ans}$$

ooo
(ii)

a_1, a_2, a_3, a_4
 $3, 3, 3, 3 \rightarrow \text{AP?}$

$$\left. \begin{aligned} d_1 &= a_2 - a_1 = 3 - 3 = 0 \\ d_2 &= a_3 - a_2 = 3 - 3 = 0 \\ d_3 &= a_4 - a_3 = 3 - 3 = 0 \end{aligned} \right\}$$

Since $d_1 = d_2 = d_3$

Yes it's AP.

$$\boxed{d = 0}$$

~~Q. 10~~ iii. $\overset{a_1}{1}, \overset{a_2}{1}, \overset{a_3}{1}, \overset{a_4}{1}, \dots$
 $\underset{2}{1}, \underset{4}{1}, \underset{6}{1}, \underset{8}{1}, \dots$

$$d_1 = a_2 - a_1 = \frac{1}{4} - \frac{1}{2} = \frac{1-2}{4} \Rightarrow \left(-\frac{1}{4}\right)$$

$$d_2 = a_3 - a_2 = \frac{1}{6} - \frac{1}{4} \Rightarrow \frac{2-3}{12} \Rightarrow \left(-\frac{1}{12}\right)$$

$$d_1 \neq d_2$$

Not an AP!

iv. $\overset{a_1}{p}, \overset{a_2}{p+90}, \overset{a_3}{p+180}, \overset{a_4}{p+270}, \dots$ where $p = (999)^{999}$

$$d_1 = a_2 - a_1 \Rightarrow \cancel{p} + 90 - \cancel{p} \Rightarrow \boxed{90}$$

$$d_2 = a_3 - a_2 = (p+180) - (p+90) \Rightarrow \cancel{p} + 180 - \cancel{p} - 90 \Rightarrow \boxed{90}$$

$$d_3 = a_4 - a_3 = (p+270) - (p+180) \Rightarrow \cancel{p} + 270 - \cancel{p} - 180 \Rightarrow \boxed{90}$$

$$\text{Since } d_1 = d_2 = d_3$$

Yes AP

$$\boxed{d=90} \quad \text{A}$$

Q.P.: A sequence is given as 1, 3, 5, 7, 9 - - -

(i) Check whether the sequence is AP or not? Yes $d_1 = d_2 = d_3 = d_4 =$

(ii) If yes find the 1001th & 1002th term?

(i) $a_{1001} = a + (1001 - 1)d$

$$= 1 + (1000)(2)$$

$$= 1 + 2000$$

$$\boxed{a_{1001} = 2001}$$

$$a_{1002} = a + (1002 - 1)d$$

$$+ 2 \Rightarrow \boxed{2003} \checkmark$$

$d = 2$
 $a = 1$

✓ $d = \text{अंतर} - \text{पिछे}$

✓ AP में d समान? $\rightarrow d$ same

✓ $a_n = a + (n-1)d$

THANK YOU COODIESS !!

Kal $\rightarrow$ Quad Ex 2 - P(1)

